

LZW Compression Challenge

Input Sequence (Length 12):

A	G	C	A	C	A	A	A	G	G	C	C
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Task Requirements:

- 1. **Trace the LZW algorithm** for the input above.
- 2. **Dictionary Constraint:** The initial dictionary contains A(0), C(1), T(2), G(3).
- 3. **Show your work:** List every step where a new dictionary entry is created.
- 4. **Calculate Efficiency:** Compare the 8-bit ASCII size vs. LZW size (using 4-bit codes).

Examiner Solution Key

Current w	Next c	New Entry (w+c)	Entry Index	Output Code
A	G	AG	4	0
G	C	GC	5	3
C	A	CA	6	1
A	C	AC	7	0
CA	A	CAA	8	6
A	A	AA	9	0
AG	G	AGG	10	4
GC	C	GCC	11	5
C	EOF	-	-	1

Final Statistics:

Final Dictionary Size: 12
Encoded Sequence: [0, 3, 1, 0, 6, 0, 4, 5, 1]
Original Size: 96 bits
Compressed Size: 36 bits (9 codes * 4 bits)
Compression Savings: 62.5%